

課題について : eax レジスタの値を求めよ processor 32 bit である

| Command            | action         | eax        | ebx        | ecx        |
|--------------------|----------------|------------|------------|------------|
| 0x8048628          |                |            |            |            |
| Mov eax, 0x9       | eax= 0x9       | 0x00000009 | -          | -          |
| Lea ebx, [eax+0x3] | Ebx =eax +3=12 | 0x00000009 | 0x0000000C | -          |
| Shl al, 0x2        | al=al<<2       | 0x00000024 | 0x0000000C | -          |
| Xor bl, al         | bl = bl XOR al | 0x00000024 | 0x00000028 | -          |
| Mov ecx, ebx       | ecx = ebx      | 0x00000024 | 0x00000028 | 0x00000028 |
| Sub ecx, eax       | ecx =ecx-eax   | 0x00000024 | 0x00000028 | 0x00000004 |
| Dec ecx            | ecx = eax-1    | 0x00000024 | 0x00000028 | 0x00000003 |
| Imul eax, ecx      | eax = eax*ecx  | 0x0000006c | 0x00000028 | 0x00000003 |

- 変更なし

1. mov eax, 0x9

eax: 0x00000009 (9) ebx と ecx 変更なし

2.lea ebx, (eax+0x3), eax: 0x00000009 (9) , ebx:  $eax + 3 = 9 + 3 = 12$  - 0x0000000C

3. shl al, 0x2 eax:

first 0x00000009 AL = 0x09 (00001001<sub>2</sub>) - 2 binary

AL << 2 00100100<sub>2</sub> = 0x24 – 16 binary xor bl, al. {eax: 0x00000024 (36)}

4. ebx:

first 0x0000000C BL = 0x0C (00001100<sub>2</sub>)

BL XOR AL: AL = 0x24 – 16binary (00100100)<sub>2</sub>

BL = 0x0C 00001100<sub>2</sub>

XOR: 00100100<sub>2</sub> ⊕ 00001100<sub>2</sub> = 00101000<sub>2</sub> = 0x28(16 binary)

5. mov ecx, ebx

eax: 0x00000024 (36)

ebx: 0x00000028 (40)

ecx , ebx- 0x00000028 (40)

6. Subtract:  $ecx - eax = 40 - 36 = 4$  ;  $0x28 - 0x24 = 0x04$ . {0x00000004 (4)}

7. Initial: 0x00000004 (4) Decrement:  $4 - 1 = 3$  0x00000003 (3)

8. imul eax, ecx

Initial: 0x00000024 (36)

Multiplication:  $eax * ecx = 36 * 3 = 108$  (10 binary ) 0x6C (16 binary)

Result:

eax: 0x0000006C (16 binary ) 108 ( 10 binary)

ebx : 0x00000028 (16 binary ) 40 ( 10 binary)

ecx : 0x00000003 (16 binary ) 3 ( 10 binary)